INCH POUND
MIL-PRF-1/22J
29 May 1998
SUPERSEDING
MIL-E-1/22H
5 March 1973

PERFORMANCE SPECIFICATION SHEET

# ELECTRON TUBE, GAS SWITCHING
# TYPES 1B35A, 1B37A, 6162, AND 5864

This specification is approved for use by all Depart
ments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tubes described herein shall consist of this document and the latest issue of Specification MIL-PRF-1.

DESCRIPTION: ATR, frequency: 1B35A - - - Fo = 9,300 MHz, incident power 250 kw
1B37A - - - Fo = 8,750 MHz
6162 - - - Fo = 9,080 MHz
5864 - - - Fo = 9,375 MHz

ABSOLUTE RATINGS:

| Parameter: | Incident power | Du | TA | Alt |
|---|---|---|---|---|
| Unit: | kw | - - - | °C | ft |
| Maximum | 250 | 0.001 | +100 | 10,000 |
| Minimum: | 4 | - - - | -40 | - - - |

(See note 1) OF Table I

PHYSICAL CHARACTERISTICS: See figure 1

TEST CONDITIONS: See note 2

| Parameter: | F | tp | prr | Du | Incident power | σ' |
|---|---|---|---|---|---|---|
| Unit: | MHz | μs | pps | - - - | kw | - - - |
| Test condition 1: | Fo ± 3% | 1.0 ± 0.1 | 1,000 | 0.001 | 200 | 1.2 (max) |
| Test condition 2: | Fo ± 3% | 1.0 ± 0.1 | 1,000 | - - - | 20 | 1.03 |

GENERAL:

Qualification - Required


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited

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Table I. Testing and Inspection.

| Inspection | Method | Notes | Conditions | Symbol | Limits Min | Limits Max | Unit |
|---|---|---|---|---|---|---|---|
| Qualification inspection | | | | | | | |
| Degradation due to vibration | 4021 | - | | - - - | - - - | - - - | - - - |
| Loaded Q (ATR tubes) | 4461 | - | | QL | - - - | 6.5 | - - - |
| High-level VSWR | 4474 | - | Test condition 2 | $\sigma$ | - - - | 1.10 | - - - |
| Conformance inspection, part 1 | | | | | | | |
| Tuning susceptance | 4482 | - | Fo ± 0.1% MHz | b = B/Yo | - - - | ± 0.06 | - - - |
| Firing time | 4486 | 3 | | t | - - - | 10 | sec |
| Arc loss | 4488 | 3 | | La | - - - | 0.8 | dB |
| Conformance inspection, part 2 | | | | | | | |
| Dielectric material strain | 4101 | - | | - - - | - - - | - - - | - - - |
| Normalized conductance | 4484 | - | Fo ± 0.1% MHz | g = G/Yo | - - - | 0.1 | - - - |
| Recovery time | 4471 | - | Test condition 1 | t | - - - | 10 | µs |
| Conformance inspection, part 3 | | | | | | | |
| Life test | - - - | - | Group D; test condition 1 | | | | |
| Type 1B35A | | - | | - - - | 500 | - - - | Hrs |
| Type 1B37A | | - | | - - - | 500 | - - - | Hrs |
| Type 6162 | | - | | - - - | 500 | - - - | Hrs |
| Type 5864 | | - | | - - - | 1,000 | - - - | Hrs |
| Life-test end points: | - - - | | | | | | |
| Arc loss | 4488 | 3 | | La | - - - | 1.0 | dB |
| Normalized conductance | 4484 | - | Fo ± 0.1% MHz | g = G/Yo | - - - | 0.1 | - - - |
| Tuning susceptance | 4482 | - | Fo ± 0.1% MHz | b = B/Yo | - - - | ± 0.06 | - - - |
| Recovery time | 4471 | - | Test condition 1; VSWR > 10 dB | t | - - - | 20 | µs |
| Temperature cycling (nonoperating) | 1027 | - | | - - - | - - - | - - - | - - - |
| Periodic-check tests | | | | | | | |
| Temperature cycling life-test end point | 1027 | 3 | Group D; 10 cycles (min) | - - - | - - - | - - - | - - - |

NOTES:
1. For 500 hours life, the duty cycle is 0.001 at pulse width of 1.0 µs. Operation at greater than 1.0 µs pulse width at 250 kw is permissible but may result in decreased life.

2. For electrical tests, the tubes shall be mounted as shown on figure 2. A suitable gasket shall be placed between the tube and the tube seat.

3. This test shall be performed at least 168 hours after pumping, and at least 24 hours after any previous discharge.



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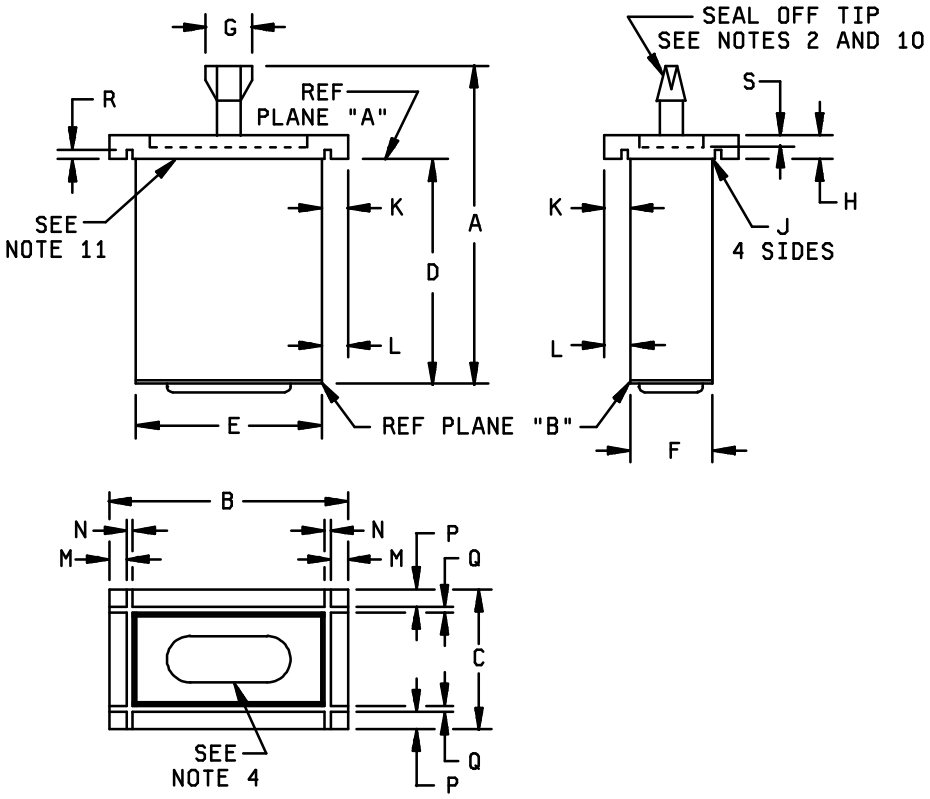


| Dimensions in inches with metric equivalents (mm) in parentheses. | | | | | |
|---|---|---|---|---|---|
| Ltr | Minimum | Maximum | Ltr | Minimum | Maximum |
| Conformance inspection, part 1 | | | | | |
| B | 1.297 (32.94) | 1.303 (33.10) | K | 0.142 (3.61) | |
| C | 0.797 (20.24) | 0.803 (20.40) | L | 0.138 (3.53) | |
| D | 1.296 (32.92) | 1.302 (33.07) | S | | 0.078 (1.98) |
| J | | 0.020 (0.51) | | | |
| Dim M through R for 1B37A only | | | | | |
| M | 0.103 (2.62) | 0.111 (2.82) | Q | 0.030 (.76) | 0.040 (1.02) |
| N | 0.030 (0.76) | 0.040 (1.02) | R | 0.035 (.89) | 0.045 (1.14) |
| P | 0.103 (2.62) | 0.111 (2.82) | | | |
| Conformance inspection, part 2 | | | | | |
| A | | 1.813 (46.05) | G | | 0.250 (6.35) |
| E | 0.990 (25.15) | 1.010 (25.65) | H | 0.117 (2.97) | 0.133 (3.38) |
| F | 0.490 (12.45) | 0.510 (12.95) | | | |

FIGURE 1. Outline drawing of electron tube types 1B35A , 1B37A, 6162 and 5864



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NOTES:

1. Metric equivalents (to the nearest 0.01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

2. Tubulation shall lie within a circle 0.38 (9.65mm) diameter maximum located from centerlines of flange. Qualification inspection required.

3. Entire tube shall be plated 100 MSI minimum silver, or equivalent. Qualification inspection required.

4. Centerlines of window shall coincide with corresponding centerlines of tube body within .015 (.38) measured in plane of window. Conformance inspection, part 2, required.

5. Two gaskets as specified on Drawing 191-JAN shall be supplied unmounted. Conformance inspection, part 2 required.

6. Dimension K shall be measured .12 (3.05 mm) from reference plane A.

7. Dimension L shall be measured .12 (3.05 mm) from window plane.

8. Reference plane A shall be normal to reference plane B.

9. Flat end plate and dimpled window optional.

10. Seal-off cap

11. Position of gasket when mounted.

FIGURE 1. Outline drawing of electron tube types 1B35A , 1B37A, 6162 and 5864. -Continued.



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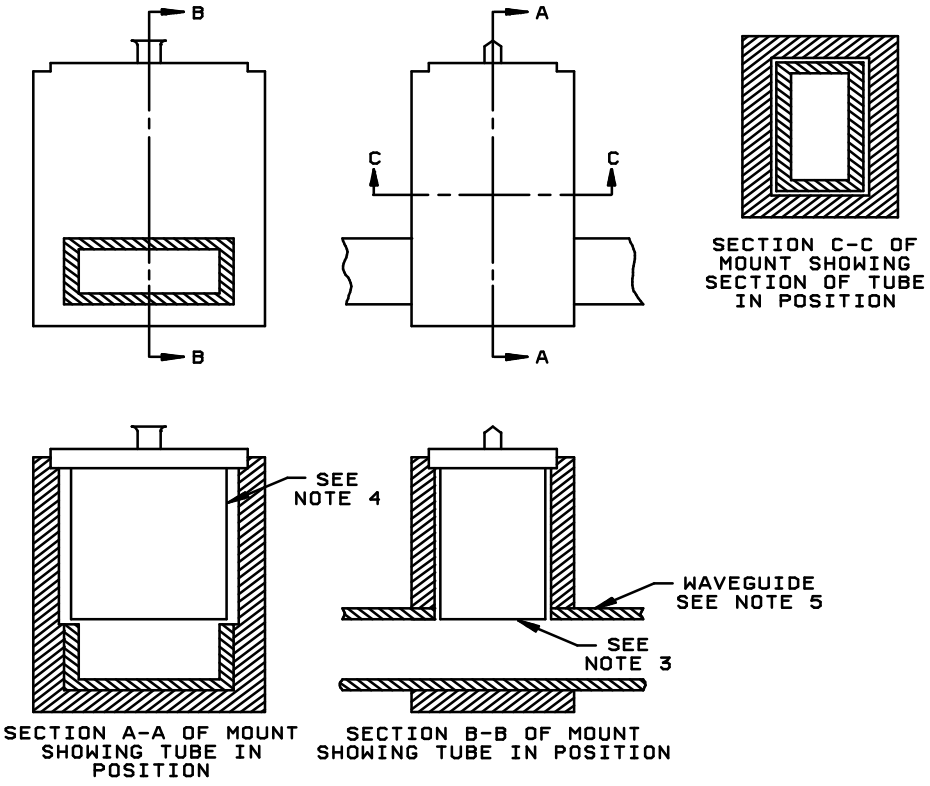


NOTES:

1. Design of holder optional.
2. Tube shall be held firmly in position by means of spring or screw clamps.
3. Bottom of the tube shall be flush with inside surface of waveguide within ± 0.004.
4. Clearance between tube walls and inner surface of mount shall be 0.030 to 0.040.
5. Waveguide shall be RG-52/U for the 1B35A, 1B37A, and 6162; and RG-51/U for the 5864. It is recommended that the RG-52/U waveguide be pressurized to prevent pulse power breakdown at 250 kw. The RG-52/U and RG-51/U waveguides shall be in accordance with MIL-W-85/1.

FIGURE 2. Test mount.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MU
Navy - AS, CG, MC, OS, SH
Air Force - 11, 99

Preparing activity:
DLA - CC

(Project 5960-3436)



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